

Maize stalk borers

Busseola fusca; *Chilo* and *Sesamia* species



Maize stalk borer larvae and frass inside a young maize tassel (Photo: CIMMYT, Flickr)



Larva feeding on maize stem (Photo: International Institute of Tropical Agriculture, Flickr)



Busseola fusca (Francis Nduati, Kenya)

Prevention

- Use resistant or tolerant varieties if available.
- Apply nitrogen, either as a mineral fertilizer, or as manure or compost, to enhance the crop's ability to sustain an attack.
- A 'push-pull' system can be implemented in which Desmodium, a repellent plant, and Napier grass, a trap crop, are intercropped with maize to push and pull the insect away from the maize (see Plantwise Factsheet for Farmers):
- An additional benefit of this system is that Desmodium is a legume that fixes nitrogen into the soil; it also acts as a ground cover that suppresses *Striga*, a parasitic weed.
- Disadvantages of this system include the space taken up by the Napier grass; the cost and lack of availability of Desmodium seed; and the difficulty in establishing the Desmodium crop.
- Intercrop with non-host plants, such as cowpeas or cassava. Adult moths will lay eggs on the non-host plants, but the larvae are unable to feed on them and will die.
- Rotate maize with a non-host plant, such as a legume (e.g. groundnut), to increase the nitrogen in the soil. This will make the next maize crop hardier and less susceptible to an attack and break the cycle of the stalk borer.

Monitoring

- Additional relevant crops: sorghum, bulrush millet, sugarcane and rice.
- Immediately after emergence, start monitoring and continue at weekly intervals until maize reaches physiological maturity.
- Consider control if you see one or two infested plants.
- Stems: Feeding damage resulting in reduction of grain production. Stems are weakened and break.
- Leaves: An early sign is small holes in straight lines or small dark larvae in the leaf funnel of young leaves.
- Central leaves become dry and withered when larvae attacks growing points, forming 'dead heart' (drying of the central shoot). The top of the plant wilts, turns yellow, and eventually dries out and dies.
- Larvae droppings (frass) are often visible on the leaves and in the stems.
- Eggs: Round, flat on top, creamy-yellow in colour. May get darker as they develop. About 1 mm in diameter.
- Larvae: Creamy-white. Might have a grey or sometimes pink colour and the head may be dark brown. Might have dark conspicuous spots and four purple stripes lengthwise across the back. Grow up to 40 mm long.
- Pupae: up to 25 mm long and shiny yellow-brown to dark brown/red-brown in colour.
- Adults/moths: Wing-span is about 20-33 mm. The forewings are light to dark brown or yellow-brown, possibly with dark patterns, and the hindwings are white to greyish-brown.
- If plants show symptoms, cut open the stem and look for larvae, pupae and frass.

Direct Control

- Dispose of crop residues after harvest to reduce stalk borer populations and limit the pest the following season:
- The crop residues can be burned, used as feed for livestock, or left on the ground exposed to the sun's heat for one month to kill the larvae and pupae.
- Apply ash or dry soil, by putting one teaspoon of ash or soil into the leaf funnel of young plants. Too much soil or ash can damage the leaf. Ash or soil should be used before the appearance of any symptoms. This means when stalk borers are known from the area or the last cropping season.
- Apply 25-50 g of neem cake powder diluted per litre water every 10 days until maize flowers. The spray should be applied into the leaf funnel every 10-14 days until flowering.
- Spray with neem seed liquid (2 handfuls of neem powder put in 10 litres of water left overnight and sieved before spraying).

Note: Pesticides may be available to control this pest. Please check with the Ministry of Agriculture in your country to find out which pesticides are registered in your country and the local restrictions for their use.